Syllabus - *Last edit* *Nov 14, 2023*

PHYS303 - Statistical Physics, Fall 2023

(aka Thermal Physics, Statistical Mechanics, Thermodynamics)

Contents

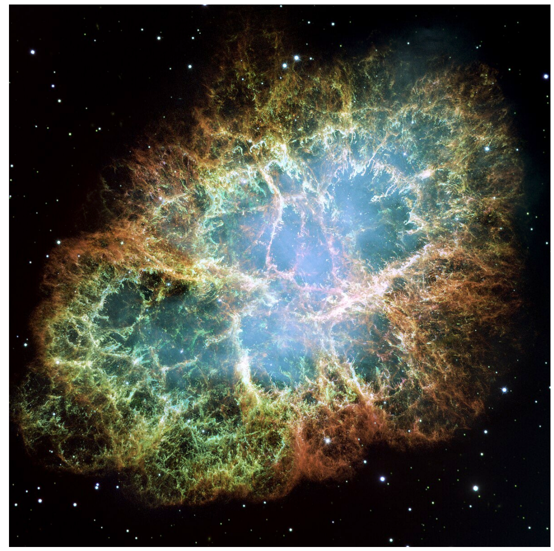

Image: The Crab Nebula. Because it's beautiful and also because concepts in Statistical Physics are used to explain the different emission features (shown by different colors) as a function of temperature and density of the atoms in the cloud.

Logistics

MWF 9.30-10.30am Hilles 108 Student drop-in hours Friday 10.30-noon. Zubrow Common	Karen Masters Office: Observatory A Email: klmasters@haverford.edu

Overview

Statistical Physics (sometimes Statistical Mechanics) is often not well understood as a topic to beginning physics students, but ends up containing many people's favorite undergraduate physics content. It combines ideas from the fields of quantum mechanics, classical mechanics, probability theory, and relativity, to name a few. It connects macroscopic, bulk properties (like temperature) to properties on a microscopic scale (like particle velocities). It is subtle and complex, and it cannot be learned by simply doing problem sets. You will need to read (books and online) talk to your classmates and to me (and anyone else who may be interested). You will directly benefit from every minute spent synthesizing the material. The more connections you make, the more likely you will fall in love with statistical mechanics. Applications range from solid state physics, to astrophysics, to information theory, to engines, to non-linear dynamics, to thunder, to biological systems, to climate change, etc. etc. There is no way to be bored in this class once you find the reason that statistical physics is important to your interests in physics or astrophysics. It's used in an astonishing array of applications of physics.

This course is not for the faint of heart! It may be difficult and time consuming at times. However, by the end of the semester, you will have a newfound appreciation for our Universe and this understanding will be a beautiful addition to your physics education.

Textbook

Concepts in Thermal Physics, 2nd Ed., by Stephen J Blundell & Katherine M Blundell

(note that the edition is important, as there were some fairly big changes from the 1st edition).

Moodle Use

This class will make use of Moodle. Assignments, any additional course information, videos, collective notes, etc. will be posted there. Please check it regularly.

Gradescope

This class will use a tool called "Gradescope" to collect assignments which will be turned in (as scanned PDFs). We will have a practice assignment the first week of class.

Slack Workspace

This course will make use of a Slack channel in the Dept. of Physics & Astronomy Slack workspace. If you are not already in the Slack workspace please reach out for an invite link.

Slack can be accessed simply using a website if you like, but it also has an app (for various phones/desktops) you can download. I find the app useful since you can get notifications when there's a new post. You can also set up to get emailed when you have Slack notifications. And you can also send private, direct messages using this platform. This is my preferred mode for quick communication.

Schedule

The aim of this class is for you to learn statistical mechanics and thermodynamics to your bones. The list of topics in the Schedule serve as a general outline for the course. As the semester progresses, it might make more sense to reorganize, add or omit various topics. I hope we will have time to cover some "Advanced Topics" as well! In the last few days of classes, you will each (likely in groups) teach a topic of your choice.

Expectations for Homework

First draft HW (submit in Gradescope):

- Always OK to use a Table of Integrals (refer to it)
- Usually OK to use Mathematica (but often not helpful; and say you used it).
  - Some problems with be marked as Mathematica not allowed
  - No screenshots from Mathematica - use it as a reference or a calculator
- Will be graded on the presentation/explanation of the answer as well as the answer:
  - You need to demonstrate your understanding of the problem solving, explain in your own words (after working on it with your group)
  - Should be able to understand the question from what you turn in as a solution (either directly including the problem, or explaining it in your own words is fine)
  - Results should be presented in a physically meaningful way (think about sensible units, number of significant figures, how numbers are presented). Reflect on the result.
- Check you actually answer all the questions in each part.

Pretty Copy (submit in Moodle):

- Graded on completion only - you have to submit something
- You are expected to look at the feedback in the first draft version and fix any mistakes
- For your own sake (as you can use these in exams), make it clear and legible.

Deadlines

The way the class is set up (see the weekly schedule) means we need to draw a firm deadline for turning in first reading quizzes (usually Mon morning) and first draft solutions (usually Fri afternoon). If there are circumstances preventing you from meeting these deadlines, please let me know ASAP. Communication is key!

Deadlines for the pretty copy solutions are more flexible, but I urge you to keep up with the schedule or you will regret it when it comes time for the mid-term and final.

Assessment (grading)

Reading Quizzes (completion - you get one for free after that this is the fraction you submit)	10%
First draft solutions (graded)	25%
Self evaluation of first draft (completion - one for free)	5%
Pretty copy solutions (completion - one for free)	20%
Presentation of advanced topic (graded)	10%
Exams - midterm and final (grade from best of two)	30%

I used the following % to 4.0-scale conversion and will aim to construct grading rubrics (for the first drafts, exams and presentations) which reflect this. I ask for your trust that grading will be done fairly to the best of my, and the graders' ability.

- 4.0 - 93%+
- 3.7 - 90-92%
- 3.3 - 87-89%
- 3.0 - 83-86%
- 2.7 - 80-82%
- 2.3 - 77-79%
- 2.0 - 73-76%
- 1.7 - 70-72%
- 1.3 - 67-69%
- 1.0 - 63-66%

Exams

There will be a midterm (right before Thanksgiving) and a final (during Finals week). Having two exams is intended as a way to provide a lower stress experience - you can try things out in the midterm in a low stress environment (it doesn't have to count) and if you do well you can have a low stress final.

For both exams you will be allowed to use the textbook and your Pretty Copy Solutions that you built over the semester. You may not use the internet (including LLM) or discuss either exam with anyone else in the class. If you are consistent and do the work for this class, the exams should be your chance to shine!

Your midterm grade will be dropped in favour of the final exam grade, unless the latter is lower - in this way your midterm result will form a baseline for the final exam grade (e.g. if you get 80% in the midterm that will be the lowest you can get for the exams, even if you completely mess up the final - if you don't do the midterm then that's on you and you get the final exam grade entered).

Presentation

During the last few classes you will get a chance to teach the class! You will need to prepare a lecture and propose a problem - this will likely need to be done in groups given the size of the class. Like anything else, public speaking requires practice. You may choose a topic that interests you from the "Advanced Topics" sections or other sections we haven't covered in class. These are all covered in our textbook, using the language and notation we're used to, but you are encouraged to research other sources as well. After you've selected a topic we can chat about the appropriate scope for your talk. The time limit will depend on the number of students in the class, and how we end up grouping. More details will come closer to the end of the semester.

Getting Help

The easiest way to fail in upper level physics is not to seek out help when you need it, and to avoid seeking help if you fall behind in the schedule. I encourage the use of peer tutors, and also study-groups for peer-support in this class, and I intend to reach out early if I see you missing assignments/deadlines.

Student drop-in hours are times I specifically set aside for you! There is no need to make an appointment. Please come and ask any questions you have. Appointments are available with me at other times via Calendly.

Slack Direct Message is the best way to get in touch with me and email is a close second best.

The Office of Academic Resources (OAR) is positioned to help you navigate Haverford from customs to commencement. Our services are free and unlimited. Our office is located on the first floor of Stokes and you're welcome anytime to study, relax, caffeinate, and connect with peers and/or a kind staff committed to your success and wellbeing. We offer an array of services, workshops, academic accountability groups, themed seasonal study breaks, and other events throughout the academic year, but our two main academic support structures are:

- **Academic Coaching:** Work with a professional, or peer, coach to set goals and develop a plan for success (however you define it). We facilitate conversations to explore 'how' to do college, from balancing time management goals, managing large, or competing, projects, developing discipline-specific study strategies, such as methods to engage and balance readings, and preparing for an exam, along with much more! Trust the process.

Please visit haverford.edu/oar to learn more about us and/or schedule an appointment. We look forward to supporting you, at any point in your academic journey.

Honor Code

Collaboration is an important part of science. You are strongly encouraged to work together and/or consult one another for work in this class. You are encouraged to consult any books necessary as well as resources on the internet. You must, however, turn in your own individual assignments/project reports, and this must be written on your own. **Copying and pasting (even parts of sentences, and including from LLM) is not permitted and is a violation of the Honour Code.** Good collaboration involves everyone understanding what is going on. Therefore even if the basic solution is shared you must explain it in your own words (including mathematical words). Please list any students that you collaborated with. Please pay attention to your classmates to make sure no one is being left out of collaborative work.

Be aware that **LLM (like ChatGPT) are often wrong** in subtle details of answers they give to physics questions (and honestly more generally). They are designed to create realistic sounding answers, which are not necessarily factually correct. While I think it is fine to use tools like ChatGPT as part of your brainstorming process, and to do things like check your grammar, code, any copying directly from such tools is plagiarism (in fact arguably ChatGPT itself plagiarizes, as it does not properly credit where it obtains the information it shares) so I plan to trust that you will not do this. In some cases use of these tools will be detrimental to your learning process. **The process and struggle to think of an answer is a significant part of learning**, and while it might be more comfortable, you do yourself a disservice if you try to skip the necessary thinking.

Inclusion, Diversity, Equity and Accessibility

Our classroom should be a place where all members will be treated with respect. We welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability - and other visible and non-visible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. If something was said in class (by anyone including myself) that made you feel uncomfortable, please talk to me about it (anonymous feedback is always an option - there will be a link in Moodle). I appreciate any opportunity to continue my learning about diverse perspectives.

In an ideal world, science would be objective. However, science is done by people, and is historically built on a small subset of privileged voices. In this class, we will make an effort to read work from a diverse group of scientists, but limits still exist on this diversity. I believe that integrating a diverse set of experiences is important for a more comprehensive understanding of science. We may discuss issues of diversity in physics/astrophysics as part of the course from time to time. Please contact me (in person or electronically) or submit anonymous feedback if you have any suggestions to improve the quality of the course materials.

Accommodations

I am committed to partnering with you on your academic and intellectual journey and recognize that you bring many strengths, perspectives and strategies as you navigate this journey. I encourage you to think proactively and strategically about leveraging these strengths, in partnership with the many resources on campus. These resources include CAPS (free and unlimited counseling is available), Office of Academic Resources, Writing Center, Student Diversity Equity and Access Team, Health Services, Professional Health Advocate, Religious and Spiritual Life, the GRASE Center, and the Advising Deans. At times you may experience challenges or stressors that impact your ability to fully engage intellectually. If the stressors are academic, I welcome the opportunity to discuss and address those stressors with you in order to find solutions together. If you are experiencing challenges or questions related to emotional health, finances, physical health, relationships, learning strategies or differences, or other related topics, I hope you will consider reaching out to the many resources here on campus. Additional information can be found at https://www.haverford.edu/deans-office-student-life/offices-resources.

Additionally, Haverford College is committed to creating a learning environment that meets the needs of its diverse student body and provides equitable access to students with disabilities. If you have (or think you may have) a disability related to mental health, chronic health, neurological state, and/or physical condition – please contact the Office of Access and Disability Services (ADS) at hc-ads@haverford.edu. It is never too late to request ADA accommodations – our bodies and circumstances are continuously changing. Please know that all inquiries and health-related information is handled in a sensitive and confidential manner.

Students who have already been approved to receive academic ADA accommodations and want to use these in this course should share their accommodation letter and make arrangements to meet with me as soon as possible to discuss how their accommodations will be implemented in this course. Please note that accommodations are not retroactive and require advance notice in order to successfully implement.

If, at any point in the semester, a disability or personal circumstances affect your learning in this course, please do not hesitate to reach out to me. I want to be sure you are aware of the full range of resources and options available to you.

It is a state law in Pennsylvania that individuals must be given advance notice that they may be recorded. Therefore, any student who has a disability-related need to audio record this class must first be approved for this ADA accommodation by Access and Disability Services and then must communicate approval to me. I will then make a general announcement to the class that audio recording may occur while respecting students' right to privacy by not identifying the individual(s).

PHYS303 - description of weekly schedule

Last edited Dec 11, 2023

Mondays - New topic (student presentation of reading)

Before Class:
Reading: Reading assignments are due before each Monday Class as listed in the Schedule. It is required that you *read* the chapter before class and *take notes*! As you read, take the time to stop and think, what does this mean? Do I really understand this step? Could I explain it to my classmates if they are confused? This reading should be your main focus before class. Your notes do not have to be beautiful, just functional and thoughtful.

Reading Quiz: On Moodle there is a quiz with simple questions asking whether or not you have done the reading (complete, partial, none) and taken notes (yes, sort of, no), and asking you for a comment and a question. These will close (I will record completion) ~1 hour before class and will help guide how I conduct the class.

In Class:
Instructor Context: I will not give a lecture that repeats what is in the book. I will either post a video that *briefly* summarizes the major points from the reading and/or give a brief summary in class. The purpose of this is to place the topics covered in the reading in a big picture context.

Student Summary: I will use your answers to the reading quiz to also ask a few people to share their notes and to highlight a few important points from the chapter. Therefore, you should come prepared to give a *very* short (no more than a few minutes), concise summary of the material. It is well known that if you prepare to present a topic you will learn it much more deeply. It is also super informative to see how each person takes notes - there's no right way, but we can take tips from each other! Come prepared to show your notes on screen (e.g have them in a Google slide, or Google Doc) or something you can email to me.

Last week's Problem Review: a review of draft solutions from last week's problem set. For each problem, I will select a first draft solution (or two) to review "in front of the class". This may be a mixture of live class time, and "posted video". As I go through each solution, I will mark it up and discuss strategies. If these are posted as videos, they will stop being available at the end of classes (i.e. they will not be available during finals week).

Wednesdays - Problem Solving

Before Class:

Problem Sets**:** Each week's readings has an associated problem set listed under PROBLEMS in the Schedule. Take some time to look at these before you get to class, but you do not need to complete them. However, you should think about them and jot down a framework for solving each problem.

In Class:

Problem Set Discussion**:** We will have breakout groups where you can discuss problem solving strategies. I will assign a "presenter" to one person in each group. When the full class reconvenes on Friday the presenters should be ready to summarize their group's thoughts and then the full class will discuss the proposed strategies. There is no single way to solve many of the problems in this class! Therefore, preparation is crucial. I expect this to be a lively process!

Fridays - Review, Cement Learning and Finish First Draft Solutions

Before Class:

Self reflection on Understanding Quiz: There will be a Moodle quiz asking for your self reflection on how well you understand the topic this week due before the Friday class. I will use these to pick up on specific topics to review in class.

In Class:

Problem Set Presentation**:** Presenters from Wednesday class will summarize their group's thoughts and then the full class will discuss the proposed strategies.

In/shortly after Class:

Complete First Draft Solutions**:** These sets should be considered your first draft and are typically due only a few hours after this class discussion. This way it will be fresh in your mind and open up the time for the preparation for the next class meeting. Please submit them in Gradescope at the end of the day. On most Fridays there will be time in class to work on first draft solutions, leaving you with time outside of class for your reading for Monday's class.

The first drafts are graded, and you will get feedback (in Gradescope).

Each first draft submission will also have a self-evaluation, where for each question you will, on a scale of 0-2, rate how well you think you did on each of the problems. **Put this at the end of your submission.** This step hopefully allows you to step outside yourself and gain perspective. You will also see - how do others perceive your work? This is a great practice. There can sometimes be a huge discrepancy between what an instructor thinks of your performance and what you think. This should help you understand which parts of problem solving you excel at and which parts need development.

***Video/In-class Feedback*:** For each problem, I will select a first draft solution (or two) to review "in front of the class". This may be a mixture of live class time, and "posted video". As I go through each solution, I will mark it up and discuss strategies. If these are posted as videos, they will stop being available at the end of classes (i.e. they will not be available during finals week).

***Pretty Copy Solutions*:** Once you have strategized each problem on your own, discussed in class, submitted a first draft, received feedback from me via the Evaluation Form, and referred to a review video (or in-class review) you should feel like you understand every step of each problem very well. On your own, write a **Pretty Copy** of each of your solutions — show all steps, explain all steps, reference equations used, and put it in context.

These will be due one week after in class or video review. This due date will be marked clearly on Moodle. This step is important since ***you will need these for the final.*** These will be graded on completion only.

Topic Schedule - PHYS 303 - Fall 2023

Week	**Meeting Date**	**Due Date**	**Topic**	**Reading**	**Problems** (Numbering from 2nd Edition)
			Preliminaries (Part I)		
~~1~~	~~Wed 6th Sep~~		Introduction to the class		
	~~Fri 8th~~	~~Fri 8th~~	Math review problems.	Math review	Math review
~~2~~	~~Mon 11th~~		Basics and Heat	Ch 1&2	
	~~Wed 13th~~				1.1, 1.2[1], 1.4 2.2, 2.3, 2.4, 2.5
	~~Fri 15th~~	~~Fri 15th~~			
3	~~Mon 18th~~		Probability	Ch 3	
	~~Wed 20th~~				3.1, 3.3[2], 3.4, 3.5.(a)-(f)
	~~Fri 22nd~~	~~Fri 22nd~~			

[1] Also give the answer in units of Earth masses, 1 $M_{\oplus}$ = 5.972x10^{24} kg

[2] Hint: for Q3.3 you will need the Taylor expansion of e^m

Week	Meeting Date	Due Date	Topic	Reading	**Problems** (Numbering from 2nd Edition)
4	Wed 27th[3]		Temperature & Boltzmann Factor	Ch 4	
	Fri 29th	Sun 1st[4]			4.2, 4.5, 4.7, 4.8 Rewrite derivations in Sections4.4 & 4.6
Kinetic Theory of Gases (Part II)					
5	Mon 2nd Oct		Maxwell-Boltzmann Distribution	Ch 5	
	Wed 4th[5]				5.1 (mathematica is not needed or allowed), 5.2, 5.5 (demonstrate WHY it makes sense), 5.6
	Fri 6th	Fri 6th			
6	Mon 9th		Pressure Mean Free Path	Ch 6.1-6.2 Ch 8	
	Wed 11th				6.4[6], 6.5 8.4 and Rewrite derivation: Section 8.1
	Fri 13th	Fri 13th			
FALL BREAK					
Transport & Thermal Diffusion (Part III)					
7	Mon 23rd		Transport in Gases Thermal Diffusion	Ch 9.1-9.3 Ch 10.1[7]	

[3] Mon 25th is Yom Kippur. The class will not meet.
[4] Moved from Friday to Sunday (since no class Monday).
[5] KLM away in Colorado. Prof. Suzanne Amador-Kane will be in class to help with problem solving.
[6] Also find volume in cubic parsecs (pc). What is the volume (in pc) and pressure (in Pa) if the mass of the cloud is $10^6\ M_\odot$ (assuming it does not become self gravitating)? Compare these pressures to 1 atm (atmospheric pressure on the surface of Earth).
[7] Focus on understanding the theoretical context in S10.1. We will develop our understanding of the various geometries from S10.2-10.4 by *doing* problems (so remember to look back at them when doing the problems).

Week	Meeting Date	Due Date	Topic	Reading	**Problems** (Numbering from 2nd Edition)
	Wed 25th				9.2[8], 9.4, Rewrite derivation of Eq.9.38 10.1, 10.8
	Fri 27th	Fri 27th			
The 1st Law (Part IV)					
8	Mon 30th		Energy Isothermal & Adiabatic Processes	Ch 11 Ch 12.1-12.3	
	Wed 1st Nov				Questions 11.1, Rewrite Example 11.2 (starting from equations 11.7 and 11.10), 11.2, 11.4.. Question 12.2, 12.6
	Fri 3rd	Fri 3rd			
The 2nd Law (Part V)					
9	Mon 6th		Heat Engines Entropy	Ch 13.1-13.6 Ch 14	
	Wed 8th				13.1, 13.2, 13.3, 13.4, 13.5 14.4, 14.8
	Fri 10th	Fri 10th			
Thermodynamics (Part VI)					
10	Mon 13th		Thermodynamic Potentials 3rd Law	Ch 16 Ch 18	
	Wed 15th				16.2a[9], 16.3, 16.6[10] 18.1, 18.2
	Fri 17th	Fri 17th			
11	Mon 20th	4pm Wed 22nd	Midterm (topics through Ch12[11])		Midterm

[8] STP = standard temperature and pressure (0°C and 1 atm)
[9] Don't forget to write in words what each quantity on the left is describing like "Change in X with Y at constant Z". Each of these types of expansion also has a name.
[10] You may start from the result in Eqn 16.79
[11] Aka all topics you should have your "Pretty copy" solutions done for.

Week	Meeting Date	Due Date	Topic	Reading	**Problems** (Numbering from 2nd Edition)
Thanksgiving Break					
Statistical Mechanics (Part VII)					
12	Mon 27th		NO CLASS		
	Wed 29th		Equipartition of Energy Partition Functions	Ch 19 Ch 20 Ch21.1-21.4[12]	19.1, 19.5 20.1 (not last part), 20.2
	Fri 1st Dec	Fri 1st Dec			
Advanced topics (TBD)					
13	Mon 4th		Advanced topics - Photons and Phonons	Ch23 Ch24	
	Wed 6th		Advanced topics - Real Gases and Earth Atmosphere	Ch26 Ch37	23.1, 24.1, 26.1 and 37.5
	Fri 8th				
14	Mon 11th		Advanced topics - Information Theory (Part 1 and Part 2)	Ch15	
	Wed 13th		Advanced topics - Stars and Compact Objects	Ch35, 36	15.1, 15.6, 35.6, 36.5
	Fri 15th				

[12] Adding Ch21 rounds out an intro to stat mech by demonstrating how to derive properties of the ideal gas a 3rd way, so it's worth a read, but don't stress the details.